

DINESH CHILKURI

DevOps Engineer

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SUMMARY

AWS Cloud & DevOps Engineer with 4+ years of experience designing, automating, and optimizing scalable, secure, and cost-efficient cloud infrastructure across AWS environments. Strong expertise in Terraform, Kubernetes (EKS), Docker, CI/CD pipelines, Infrastructure as Code (IaC), and cloud automation, with hands-on experience managing production-grade deployments and observability platforms. Experienced in building end-to-end DevOps and MLOps workflows using GitLab CI/CD, Jenkins, Amazon SageMaker, and Bedrock, including GenAI and RAG-based applications. Passionate about improving system reliability, scalability, security, and developer productivity through automation and modern cloud engineering practices.

EXPERIENCE

Sr AWS Cloud Engineer
Cognizant

Jun 2024 - Present
Seattle, WA

- Spearheaded end-to-end project delivery as a developer, encompassing requirement gathering, design using AWS Well-Architected Framework, development, rigorous testing, and robust CI/CD implementation, while architecting cloud-based solutions across multiple AWS platforms.
- Designed, developed, and maintained reusable, versioned Terraform modules to provision secure and scalable AWS infrastructure (VPC, EC2, RDS, S3, EKS), ensuring input validation, output standardization, and seamless onboarding across dev, test, and prod environments.
- Implemented Sentinel policies in HashiCorp Terraform Enterprise to enforce organizational compliance, security, and cost governance during infrastructure provisioning, integrating policy-as-code into the IaC workflow.
- Integrated Terraform workflows with CI/CD pipelines (GitLab, Jenkins) for automated validation and deployment; managed module and policy releases using Git version control, tagging, and rollback strategies.
- Provisioned and maintained secure, cost-optimized AWS infrastructure using Terraform, preserving state integrity and isolating environments (dev/test/prod) via workspaces.
- Designed and implemented least-privilege IAM policies and roles for Terraform-based AWS provisioning, ensuring secure, compliant, and role-specific access to AWS services across multi-environment deployments.
- Led migration of critical AWS services (RDS, SQS, SNS, DynamoDB, Redshift, Glue, EFS, EC2) from legacy provisioning to Terraform Enterprise, automating resource discovery and CI/CD integration via GitLab while ensuring zero-downtime and audit-ready infrastructure.
- Designed and implemented CI/CD pipelines for Spring Boot applications using GitLab CI, Maven, and Docker; automated provisioning and deployments to AWS ECS using Terraform, with image/artifact management through Nexus and ECR.
- Built and deployed AWS Lambda functions using GitLab CI/CD by automating Python ZIP packaging, dependency resolution, and artifact uploads to Nexus and S3, enabling version-controlled deployments and triggered Lambda executions via S3 and SQS using Terraform.
- Implemented Blue/Green deployments on AWS ECS using a shared ALB and Route 53 DNS; automated traffic switching through ALB Listener rules and host-based routing to enable zero-downtime and phased rollouts, leveraging Datadog for real-time monitoring.
- Deployed Dockerized Python batch jobs to AWS using ECR and Artifactory; provisioned AWS Batch Compute Environments, Job Queues, and Job Definitions; orchestrated execution via Step Functions triggered by custom workflows.
- Built, automated, and optimized GitLab CI/CD pipelines for end-to-end software delivery, integrating build, test, security scanning with Harbor, containerization with Docker, and deployment to Kubernetes (EKS) and ECS.
- Provisioned, configured, and managed Kubernetes clusters on AWS EKS, deploying containerized microservices with Helm charts, managing cluster scaling, and monitoring using CloudWatch and Prometheus/Grafana.
- Architected and managed scalable, secure AWS cloud infrastructure, including EC2, S3, VPC, RDS, Lambda, EKS, IAM, and CloudWatch, ensuring high availability and fault tolerance.
- Architected and deployed enterprise GenAI applications using Amazon Bedrock and Claude/Llama models, implementing Retrieval-Augmented Generation (RAG) with OpenSearch for enhanced factual accuracy and reduced hallucinations in production chatbots.
- Developed autonomous AI agents using Bedrock Agents and ReAct prompting to orchestrate multi-step business logic via AWS Lambda, while optimizing LLM cost and performance through supervised fine-tuning and Provisioned Throughput.
- Engineered end-to-end MLOps pipelines using SageMaker Studio and SageMaker Pipelines, automating the full lifecycle from data preprocessing with Data Wrangler to CI/CD-driven model deployment and experiment tracking.

Principle Cloud Engineer
Synopsis

Aug 2022 - Apr 2023
India

- Engineered and deployed serverless cloud applications leveraging AWS Lambda, Azure Functions, and Google Cloud Functions, integrating S3, Blob Storage, and Google Cloud Storage for data pipelines, optimized compute utilization and reduced cloud costs by 7% through automated scaling and resource optimization.
- Optimized scalable ML infrastructure by deploying high-availability models on SageMaker Managed Endpoints, leveraging Multi-Model Endpoints (MME) and auto-scaling to maximize compute utilization and reduce hosting overhead.
- Strengthened model governance and reliability by utilizing SageMaker Model Monitor and Clarify for proactive drift detection, while fine-tuning domain-specific NLP/CV models via SageMaker JumpStart.
- Developed and productionized AI/ML workloads (NLP, Computer Vision) using AWS SageMaker, PyTorch, and Deep Learning frameworks, deploying models such as BERT, YOLOv8, and ResNet50, achieving 91%+ accuracy and improving inference latency by 14%.
- Architected hybrid real-time and streaming data pipelines using Apache Kafka, Python, and Google Cloud Pub/Sub, reducing data lag by 9%, job retries by 6%, and enhancing cross-cloud event orchestration for low-latency applications.
- Standardized Infrastructure as Code (IaC) using Terraform, AWS CloudFormation, and Azure ARM templates, automating CI/CD pipelines (GitHub Actions, Jenkins) for API deployments, eliminating configuration drift and increasing test coverage by 9%.
- Led peer code reviews and mentored cross-functional engineering teams, enforcing PEP8 standards, DevSecOps best practices, and GitOps workflows for containerized deployments with Docker, resulting in faster release cycles and improved code quality metrics by 11%.

Full Stack Engineer
PhaseZero.ai

Aug 2020 - Aug 2022
India

- Engineered scalable, cloud-native web platforms using Django, Django REST Framework (DRF), and MVT architecture, deploying modular RESTful APIs on AWS Elastic Beanstalk and Azure App Services, reducing regression bugs by 6% and improving system maintainability.
- Optimized database performance and caching strategy through PostgreSQL query tuning (select_related/prefetch_related), Redis, and AWS ElastiCache, cutting query count by 14%, enhancing p95 response times by 6%, and increasing transaction throughput.
- Implemented secure authentication and authorization using JWT, OAuth2, and Role-Based Access Control (RBAC) with integration to AWS Cognito and Azure AD, while automating CI/CD pipelines via GitHub Actions and Terraform, strengthening deployment consistency and test automation.
- Deployed and managed high-availability applications using AWS EC2, Load Balancers, NGINX, and Gunicorn, incorporating auto-scaling and health checks to increase uptime by 6%, decrease page load times by 8%, and enhance resiliency under high traffic.

EDUCATION

Master of Science (M.S.) - Data Analytics
The Catholic University of America

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Bachelor of Technology (B.Tech) – Mechanical Engineering
VJIT, India

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SKILLS

Cloud Platforms AWS (EC2, S3, VPC, RDS, DynamoDB, Lambda, API Gateway, IAM, CloudWatch, EKS, CloudFormation, EventBridge, SQS, SNS, CloudTrail, AWS Config, Aurora, SageMaker, Bedrock), Azure Kubernetes Services Operations

Infrastructure as Code (IaC) AWS CloudFormation, Ansible, Sentinel/OPA policies, Terraform

Containerization & Orchestration AWS ECS, Docker, Kubemetes (AWS EKS)

CI/CD Tools GitHub Actions, Jenkins, GitLab

SCM Tools GitHub, AWS Code Commit

Monitoring & Observability CloudWatch, Datadog, Dynatrace, ELK stack, Grafana (LGMT Stack: Loki, Mimir, Tempo, OpenTelemetry), Kibana, Pagerduty, Prometheus, Splunk

Programming & Scripting Bash (AWS CLI), Django, JSON, JavaScript (React), Python

Messaging Systems Amazon EventBridge, Amazon SNS, Amazon SQS, Kafka, Rabbit MQ

Database Management Amazon Aurora, Amazon DynamoDB, MySQL (implied via RDS/Aurora)

AI/ML Technologies Amazon SageMaker, Amazon Titan, Amazon Bedrock, Anthropic Claude

Service Mesh Conduit, Envoy, LinkerD

Collaboration Tools Confluence, Jira, Slack

General Development Source Code, Troubleshooting

CERTIFICATIONS

AWS Knowledge Compute

AWS Knowledge Cloud Essentials

AWS Knowledge Data Migration